# SAP S/4HANA Cost Center Controlling Project

## Cost Center Budget, Actual Cost and Commitment Analysis

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# 1. Project Overview

This project is a simulated business case based on **SAP S/4HANA Controlling (CO)**, with a focus on **Cost Center Accounting and Internal Order Accounting**.

The project demonstrates how an organization can use controlling concepts to plan, monitor and control costs across different operational departments.

The business scenario is designed around a manufacturing company, allowing the project to connect SAP Controlling with an operational environment involving production, warehouse, procurement, quality and logistics activities.

The project will focus primarily on **Controlling**, while using these operational areas as the business context.

# 2. Company Background

**Company Name:** ABC Manufacturing GmbH

**Industry:** Manufacturing

ABC Manufacturing GmbH manufactures industrial components and operates several departments that incur different types of costs.

Management wants better visibility into how departmental budgets are being used, what costs have already occurred, what costs are committed for the future, and which cost centers may exceed their available budgets.

To address this, the project applies SAP S/4HANA Controlling concepts to analyze **budgets, actual costs, commitments and budget availability**.

# 3. Business Problem

The company currently needs better visibility into departmental spending.

Looking only at actual costs does not provide the complete picture because some costs have already been committed through purchasing activities but have not yet become actual costs.

For example, a department may have a budget of €50,000 and actual costs of €30,000. However, if the department has an additional €15,000 in open purchase commitments, only €5,000 of the budget is effectively available.

| Item | Amount |
|---|---|
| Departmental budget | €50,000 |
| Actual costs | €30,000 |
| Open purchase commitments | €15,000 |
| **Effectively available budget** | **€5,000** |

*Table 1: Illustrative Budget Availability Example*

Therefore, management needs to consider:

- Planned budget
- Actual costs
- Commitments
- Available budget
- Budget utilization
- Budget variance

The project will demonstrate how these elements can be analyzed using SAP S/4HANA Controlling concepts.

# 4. Cost Center Structure

The company is divided into five cost centers for the purpose of this project.

| Cost Center | Department | Main Area |
|---|---|---|
| CC1000 | Production | Manufacturing operations |
| CC2000 | Warehouse | Storage and warehouse operations |
| CC3000 | Procurement | Purchasing activities |
| CC4000 | Quality | Quality inspection and control |
| CC5000 | Logistics | Transportation and distribution activities |

*Table 2: Cost Center Structure*

## Purpose of the Cost Centers

### CC1000 – Production

Used to collect and monitor costs related to manufacturing operations.

**CC2000 – Warehouse**

Used to monitor costs associated with warehouse and storage activities.

**CC3000 – Procurement**

Used to monitor costs related to purchasing and procurement activities.

**CC4000 – Quality**

Used to monitor costs related to inspection, testing and quality control activities.

**CC5000 – Logistics**

Used to monitor costs related to transportation and distribution activities.

# 5. Project Objective

The main objective of this project is to demonstrate how SAP S/4HANA Controlling concepts can be used to:

- Plan budgets for different cost centers.
- Track actual costs against the allocated budget.
- Identify future financial commitments.
- Calculate the available budget after considering actual costs and commitments.
- Identify cost centers at risk of exceeding their budgets.
- Analyze budget utilization and variances.
- Demonstrate the effect of budget transfers, supplements and returns.
- Apply Internal Order Accounting concepts within the overall controlling scenario.

# 6. Project Scope

The project will focus on the following SAP S/4HANA Controlling concepts:

- Cost Center Accounting
- Internal Order Accounting
- Planning and Budgeting
- Actual Costs
- Commitments
- Budget Availability Control

- Budget Variance Analysis
- Budget Transfers
- Budget Supplements
- Budget Returns

The project will use simulated data to demonstrate these concepts and will not represent a live SAP S/4HANA implementation.

# 7. Project Data Structure

To analyze the financial performance of the cost centers, the project will use a structured dataset based on the key concepts of SAP S/4HANA Controlling.

The data will be organized around **budgets, actual costs, commitments and internal orders**. This will allow the project to compare planned and actual spending and determine the budget that remains available for each cost center.

## 7.1 Cost Center Data

The project will use five cost centers representing the main operational departments of ABC Manufacturing GmbH:

- Production
- Warehouse
- Procurement
- Quality
- Logistics

Each cost center will have a unique identification code and will be used to assign and analyze the costs associated with the respective department.

## 7.2 Budget Data

An annual budget will be assigned to each cost center. The budget represents the amount planned or allocated for the respective department during the fiscal year.

The budget data will be used as the basis for monitoring spending and performing budget availability analysis.

## 7.3 Actual Cost Data

Actual cost data will represent costs that have already occurred during the fiscal year.

The costs will be assigned to the relevant cost center and categorized according to the type of expense, such as materials, maintenance, utilities or other operating costs.

This will allow the project to analyze how each cost center is using its allocated budget.

## 7.4 Commitment Data

The project will also include commitments representing future costs that have already been requested or ordered.

Purchase Requisitions (PRs) and Purchase Orders (POs) will be used as examples of procurement-related commitments assigned to cost centers.

The commitment data will allow the project to distinguish between costs that have already occurred and costs that are expected to occur in the future.

## 7.5 Available Budget

The available budget will be determined by considering both actual costs and commitments against the allocated budget.

For the purpose of this project:

**Available Budget = Budget − Actual Costs − Commitments**

This calculation will provide an overview of the budget remaining after considering both realized and committed spending.

## 7.6 Budget Utilization and Variance

The project will calculate budget utilization and budget variance for each cost center.

These calculations will help identify cost centers with relatively high spending levels and those that may require closer budget monitoring.

## 7.7 Budget Availability Control

A simulated Budget Availability Control scenario will be included to demonstrate how budget consumption can be monitored against defined tolerance limits.

The scenario will show how warnings or errors can be triggered when spending or commitments exceed defined budget thresholds.

### 7.8 Internal Order Data

An Internal Order will also be included in the project to demonstrate the use of Internal Order Accounting for a specific activity or project within the organization.

The Internal Order will allow costs related to a specific activity to be identified separately from the regular costs collected by the cost centers.

### 7.9 Data Analysis

The collected data will later be analyzed to compare:

- Budget against actual costs
- Budget against commitments
- Actual costs and commitments against the available budget
- Budget utilization and variance across cost centers

The results will subsequently be used to create a management-oriented dashboard and support the overall analysis of the project.

## 8. Cost Centers and Cost Types

### 8.1 Cost Center Structure

The project uses five cost centers to represent the main operational departments of ABC Manufacturing GmbH. Each cost center is used to collect and analyze costs associated with a specific area of the organization.

| Cost Center | Department | Main Area |
|---|---|---|
| CC1000 | Production | Manufacturing operations |
| CC2000 | Warehouse | Storage and warehouse operations |
| CC3000 | Procurement | Purchasing activities |
| CC4000 | Quality | Quality inspection and control |
| CC5000 | Logistics | Transportation and distribution activities |

*Table 3: Cost Center Structure Used in the Analysis*

## 8.2 Cost Types

To analyze the costs of each cost center, the project uses five main cost types: Materials, Labor, Utilities, Maintenance and Services.

| Cost Type | Description |
|---|---|
| Materials | Costs related to materials and consumables used in operations |
| Labor | Employee-related operating costs |
| Utilities | Costs such as electricity and other utilities |
| Maintenance | Costs related to maintenance and repairs |
| Services | Costs for external services purchased from suppliers |

*Table 4: Cost Type Definitions*

The relevant cost types are assigned to each cost center according to its operational activities.

| Cost Center | Applicable Cost Types |
|---|---|
| CC1000 Production | Materials, Labor, Utilities, Maintenance |
| CC2000 Warehouse | Labor, Utilities, Maintenance, Services |
| CC3000 Procurement | Labor, Services, Utilities |
| CC4000 Quality | Labor, Services, Utilities, Maintenance |
| CC5000 Logistics | Labor, Utilities, Maintenance, Services |

*Table 5: Applicable Cost Types by Cost Center*

This structure will be used later to create the simulated actual cost and commitment data. It will also allow costs to be analyzed by both cost center and cost type.

# 9. Budget Data

## 9.1 Annual Cost Center Budget

ABC Manufacturing GmbH allocates an annual budget to each cost center. The budget represents the amount planned for spending during the fiscal year and provides the basis for monitoring cost center performance.

The simulated annual budgets used in this project are:

| Cost Center | Department | Annual Budget |
|---|---|---|
| CC1000 | Production | €120,000 |
| CC2000 | Warehouse | €60,000 |
| CC3000 | Procurement | €40,000 |
| CC4000 | Quality | €50,000 |
| CC5000 | Logistics | €80,000 |
| **Total** | | **€350,000** |

*Table 6: Annual Cost Center Budget*

The budget values are simulated for the purpose of this portfolio project and do not represent actual company data.

The different budget levels are intended to represent the relative size of the operational areas within the fictional manufacturing company and to provide a basis for subsequent budget and cost analysis.

## 9.2 Budget Analysis

The annual budget will later be compared with actual costs and commitments to determine how much of the allocated budget has been used and how much remains available.

For example, if the Production cost center has an annual budget of €120,000, actual costs of €75,000 and commitments of €30,000, the available budget would be:

**Available Budget = Budget − Actual Costs − Commitments**

**Available Budget = €120,000 − €75,000 − €30,000 = €15,000**

| Item | Amount |
|---|---|
| Annual budget (Production, CC1000) | €120,000 |
| Actual costs | €75,000 |
| Commitments | €30,000 |
| **Available budget** | **€15,000** |

*Table 7: Available Budget Example for Production (CC1000)*

This calculation will be applied to all cost centers during the data analysis stage.

# 10. Actual Cost Data

## 10.1 Overview

Actual cost data represents costs that have already been incurred and recorded by the company. In this project, actual costs are assigned to individual cost centers and categorized by expense type.

The dataset is based on a simulated business scenario for ABC Manufacturing GmbH. It includes materials, labor, utilities, maintenance and services costs across the company's five cost centers.

## 10.2 Actual Cost Summary

The simulated actual costs for the five cost centers are summarized below.

| Cost Center | Department | Annual Budget | Total Actual Costs |
|---|---|---|---|
| CC1000 | Production | €120,000 | €75,000 |
| CC2000 | Warehouse | €60,000 | €30,000 |
| CC3000 | Procurement | €40,000 | €20,000 |
| CC4000 | Quality | €50,000 | €30,000 |
| CC5000 | Logistics | €80,000 | €45,000 |
| **Total** | | **€350,000** | **€200,000** |

*Table 8: Actual Cost Summary*

The total actual costs amount to €200,000, representing approximately 57.14% of the combined annual budget of €350,000.

## 10.3 Purpose of the Analysis

The actual cost dataset will be used to:

- Compare actual costs with the annual budget for each cost center.
- Identify the cost centers with the highest recorded spending.
- Analyze costs by expense category.
- Calculate budget utilization and variance.
- Combine actual costs with outstanding commitments in the next stage of the project.

The dataset is simulated for portfolio and learning purposes. It does not represent transactions extracted from or posted in a live SAP S/4HANA system.

# 11. Commitment Data

## 11.1 Overview

Commitments represent expected future spending associated with purchasing activities. In SAP S/4HANA Controlling, purchase requisitions and purchase orders can create commitments that indicate costs expected to occur in the future.

When the relevant goods or services are received and actual costs are recorded, the corresponding commitment is reduced according to the applicable process and system configuration.

This project uses simulated open purchase requisitions and purchase orders to demonstrate how commitments affect the budget position of each cost center.

## 11.2 Commitment Summary

The simulated open commitments for ABC Manufacturing GmbH are summarized below.

| Cost Center | Department | Annual Budget | Actual Costs | Open Commitments |
|---|---|---|---|---|
| CC1000 | Production | €120,000 | €75,000 | €30,000 |
| CC2000 | Warehouse | €60,000 | €30,000 | €17,000 |
| CC3000 | Procurement | €40,000 | €20,000 | €12,000 |
| CC4000 | Quality | €50,000 | €30,000 | €15,000 |
| CC5000 | Logistics | €80,000 | €45,000 | €25,000 |
| **Total** | | **€350,000** | **€200,000** | **€99,000** |

*Table 9: Commitment Summary*

The total simulated open commitments amount to €99,000.

## 11.3 Purpose of Commitment Analysis

Analyzing commitments alongside actual costs provides a more complete view of expected budget consumption.

For this project, the available budget will be calculated as:

**Available Budget = Annual Budget − Actual Costs − Open Commitments**

This calculation assumes that the commitments are open and have not already been included in actual costs.

The commitment dataset will be used in the subsequent budget availability, utilization and variance analysis.

All commitment records and document identifiers in this project are simulated and do not represent actual transactions in a live SAP S/4HANA system.

# 12. Budget Analysis

## 12.1 Objective

The objective of the budget analysis is to combine annual budgets, actual costs and open commitments to evaluate the financial position of each cost center.

The analysis uses the following formulas:

**Available Budget = Budget − Actual Costs − Commitments**

**Budget Utilization = (Actual Costs + Commitments) ÷ Budget × 100**

**Budget Variance = Budget − (Actual Costs + Commitments)**

For this project, budget variance represents the remaining budget after actual costs and open commitments have been considered.

The analysis in this section represents the initial budget position based on the original annual budget of €350,000, before the budget adjustments described in Section 15. The results provide a baseline for comparison with the revised budget analysis presented in the subsequent sections.

## 12.2 Results

| Cost Center | Department | Budget | Actual Costs | Commitments | Available Budget | Utilization |
|---|---|---|---|---|---|---|
| CC1000 | Production | €120,000 | €75,000 | €30,000 | €15,000 | 87.50% |
| CC2000 | Warehouse | €60,000 | €30,000 | €17,000 | €13,000 | 78.33% |
| CC3000 | Procurement | €40,000 | €20,000 | €12,000 | €8,000 | 80.00% |
| CC4000 | Quality | €50,000 | €30,000 | €15,000 | €5,000 | 90.00% |
| CC5000 | Logistics | €80,000 | €45,000 | €25,000 | €10,000 | 87.50% |
| **Total** | | **€350,000** | **€200,000** | **€99,000** | **€51,000** | **85.43%** |

*Table 10: Budget Analysis Results (Original Budget)*

## 12.3 Findings

The analysis indicates that Quality has the highest budget utilization at 90%, followed by Production and Logistics at 87.5%. These departments may require closer monitoring because a relatively large share of their budgets is already consumed by actual costs and commitments.

Warehouse has the lowest utilization at approximately 78.33%. Across all five cost centers, the remaining budget is €51,000, equivalent to 14.57% of the total annual budget.

These findings are based on simulated data and are intended to demonstrate how cost center budget analysis can support management decisions.

The results will be used in the next stage to demonstrate budget availability monitoring and identify departments that may approach their spending limits.

The analysis in this section represents the initial budget position based on the original annual budget of €350,000, before the budget adjustments described in Section 15. The results provide a baseline for comparison with the revised budget analysis presented in the subsequent sections.

# 13. Budget Availability Control

## 13.1 Overview

Budget Availability Control (BAC) is used in this project to demonstrate how management can monitor budget consumption and identify cost centers that are approaching or exceeding their allocated budgets.

Budget consumption is calculated using actual costs and open commitments.

**Budget Consumption = Actual Costs + Open Commitments**

**Budget Utilization = Budget Consumption ÷ Annual Budget × 100**

The project uses simulated warning and error thresholds to classify the budget position of each cost center.

## 13.2 Simulated Thresholds

The following thresholds are defined for this portfolio project:

| Status | Threshold | Interpretation |
|---|---|---|
| Normal | Below 80% | Budget consumption is below the warning threshold |
| Warning | 80% to below 100% | Management should monitor spending |
| Error | 100% or more | The budget has been fully consumed or exceeded |

*Table 11: Simulated BAC Thresholds*

These thresholds are illustrative assumptions for the simulated business case and do not represent a claim about SAP S/4HANA's default configuration.

## 13.3 Results

| Cost Center | Department | Budget Utilization | BAC Status |
|---|---|---|---|
| CC1000 | Production | 87.50% | Warning |
| CC2000 | Warehouse | 78.33% | Normal |
| CC3000 | Procurement | 80.00% | Warning |
| CC4000 | Quality | 90.00% | Warning |
| CC5000 | Logistics | 87.50% | Warning |

*Table 12: BAC Results (Original Budget)*

### 13.4 Findings

Quality has the highest budget utilization at 90%, followed by Production and Logistics at 87.5%. Procurement has reached the warning threshold of 80%, while Warehouse remains below it.

None of the cost centers has exceeded its budget in the simulated scenario.

The results demonstrate how warning thresholds can help management identify departments requiring closer monitoring before their budgets are fully consumed.

The Excel model classifies the cost centers using formulas. It simulates the analysis of budget consumption and does not represent a live SAP S/4HANA Budget Availability Control configuration.

## 14. Internal Order Accounting

### 14.1 Overview

Internal Order Accounting is used to collect and monitor costs associated with specific activities, events, or projects.

While Cost Center Accounting focuses on costs associated with organizational areas, Internal Order Accounting allows costs to be tracked for a particular activity.

This project introduces an internal order for a production equipment improvement initiative at ABC Manufacturing GmbH.

## 14.2 Internal Order Scenario

The project uses the following simulated internal order:

| Field | Value |
|---|---|
| Internal Order ID | IO1001 |
| Description | Production Equipment Improvement |
| Responsible Cost Center | CC1000 |
| Department | Production |
| Planned Budget | €15,000 |
| Actual Costs | €8,000 |
| Open Commitments | €4,000 |
| Remaining Order Budget | €3,000 |

*Table 13: Internal Order Scenario (IO1001)*

## 14.3 Remaining Order Budget Calculation

The remaining order budget is calculated as follows:

**Remaining Order Budget = Planned Budget − Actual Costs − Open Commitments**

**Remaining Order Budget = €15,000 − €8,000 − €4,000 = €3,000**

The calculation indicates that €3,000 remains available within the simulated internal order budget after accounting for actual costs and open commitments.

## 14.4 Purpose of Internal Order Analysis

This scenario demonstrates how a specific activity can be monitored separately from the general costs associated with a department.

The internal order dataset will support the analysis of planned budgets, actual costs, commitments, and remaining order budgets.

The scenario is simulated for portfolio and learning purposes. It does not represent an internal order created or processed in a live SAP S/4HANA system.

The internal order scenario is presented separately as an illustrative example of activity-based cost monitoring. Its financial figures are not added separately to the departmental actual costs and commitments used in the consolidated cost center analysis. This distinction avoids double

counting within the overall financial results.

# 15. Budget Adjustments

## 15.1 Overview

During a fiscal year, an organization may need to adjust its budget allocations in response to changing operational requirements.

This project demonstrates three budget adjustment concepts: supplements, returns and transfers.

| Adjustment Concept | Description |
|---|---|
| Budget Supplement | An increase in the budget allocation. |
| Budget Return | A reduction in the budget allocation when funds are no longer required. |
| Budget Transfer | A reallocation of budget from one cost center to another. |

*Table 14: Budget Adjustment Concepts*

The examples are simulated for ABC Manufacturing GmbH.

## 15.2 Simulated Budget Adjustments

| Adjustment ID | Adjustment Type | Cost Center | Department | Amount |
|---|---|---|---|---|
| BA001 | Supplement | CC1000 | Production | €10,000 |
| BA002 | Return | CC2000 | Warehouse | €5,000 |
| BA003 | Transfer Out | CC3000 | Procurement | €8,000 |
| BA004 | Transfer In | CC1000 | Production | €8,000 |

*Table 15: Simulated Budget Adjustments*

The transfer is recorded on both the sending and receiving cost centers to reflect the reduction and increase in their respective budget allocations.

## 15.3 Revised Budget Calculation

The revised budget is calculated as follows:

**Revised Budget = Original Budget + Supplements − Returns + Transfers In − Transfers Out**

| Cost Center | Department | Original Budget | Net Adjustment | Revised Budget |
|---|---|---|---|---|
| CC1000 | Production | €120,000 | +€18,000 | €138,000 |
| CC2000 | Warehouse | €60,000 | −€5,000 | €55,000 |
| CC3000 | Procurement | €40,000 | −€8,000 | €32,000 |
| CC4000 | Quality | €50,000 | €0 | €50,000 |
| CC5000 | Logistics | €80,000 | €0 | €80,000 |
| **Total** | | **€350,000** | **+€5,000** | **€355,000** |

*Table 16: Revised Budget by Cost Center*

The overall budget increases from €350,000 to €355,000 because the €10,000 supplement exceeds the €5,000 return by €5,000. The €8,000 transfer reallocates budget between Production and Procurement without changing the overall total.

## 15.4 Purpose of the Analysis

The budget adjustment dataset demonstrates how changes in budget allocations can be recorded and analyzed.

The revised budgets will be used in the subsequent analysis to examine how budget adjustments affect available budget, utilization and budget variance.

The examples are simulated and do not represent actual budget postings in a live SAP S/4HANA system.

## 15.5 Revised Budget Calculation

The revised budget represents the original departmental budget after accounting for the applicable budget adjustments. These adjustments include budget supplements, budget returns, and transfers between cost centers.

The revised budget is calculated using the following formula:

**Revised Budget = Original Budget + Net Budget Adjustments**

The budget adjustments are incorporated into the departmental budget figures before calculating total consumption, available budget, and budget utilization.

For this project, the original total budget was €350,000. The budget adjustments resulted in a net increase of €5,000, bringing the revised total budget to €355,000.

The revised budget provides the financial baseline for evaluating departmental expenditure and outstanding commitments.

### 15.6 Impact of Budget Adjustments on Financial Analysis

Budget adjustments influence the amount of funding available to individual departments and the organization as a whole.

A budget supplement increases the budget allocation of the relevant department, while a budget return reduces its allocation. A budget transfer reallocates funds between departments without changing the overall budget total.

These changes affect the calculation of available budget and budget utilization. Therefore, the revised budget rather than the original budget is used as the reference for the subsequent analysis.

In this project, the revised budget figures are used to compare departmental budgets with actual costs and open commitments. This approach provides a consolidated view of budget consumption and highlights departments that may require closer financial monitoring.

The analysis is based on simulated data prepared for this portfolio project and does not represent actual postings or budget adjustments in a live SAP S/4HANA system.

## 16. Cost Center Budget and Consumption Analysis

### 16.1 Introduction

Cost center budget and consumption analysis is used to evaluate how departmental financial resources are being utilized in relation to approved budgets. By comparing revised budgets with actual costs and open commitments, management can assess current spending, identify departments approaching their budget limits, and determine the amount of budget remaining.

In this project, the analysis is performed using simulated financial data for five departments within ABC Manufacturing GmbH. The revised budget serves as the reference value for measuring departmental consumption and budget availability.

The analysis considers four main indicators: revised budget, actual costs, open commitments, and total consumption. These indicators are used to calculate available budget and budget utilization for each department.

## 16.2 Revised Budget

The revised budget represents the departmental budget after incorporating the applicable budget adjustments.

The total original budget was €350,000. Following the simulated budget adjustments, the total revised budget increased to €355,000.

The revised budget is used as the financial baseline for evaluating expenditure and outstanding commitments. Using the revised budget ensures that the analysis reflects the updated allocations rather than relying solely on the original budget values.

The revised budgets differ across departments because each cost center has its own allocation and adjustment history.

## 16.3 Actual Costs

Actual costs represent expenditure recorded against the respective cost centers in the simulated dataset.

For this project, total actual costs amount to €200,000 across the five departments.

Actual costs provide an indication of the expenditure already incurred during the reporting period. However, actual costs alone do not provide a complete picture of budget consumption because additional expenditure may arise from outstanding commitments.

Therefore, actual costs are analyzed together with open commitments to provide a broader view of departmental financial exposure.

## 16.4 Open Commitments

Open commitments represent outstanding financial obligations that have not yet been fully reflected in actual costs within the simulated dataset.

Examples may include outstanding purchase orders or other committed expenditure.

The total open commitments in this project amount to €99,000.

Including commitments in the analysis helps provide a more forward-looking view of budget utilization. A department may have relatively low actual costs but still have limited budget availability if substantial commitments remain outstanding.

For this reason, open commitments are considered alongside actual costs when calculating total consumption.

## 16.5 Total Budget Consumption

Total consumption combines actual costs and open commitments to estimate how much of the revised budget has already been used or committed.

**Formula:**

Total Consumption = Actual Costs + Open Commitments

For the overall simulated company:

| Indicator | Amount |
|---|---|
| Actual Costs | €200,000 |
| Open Commitments | €99,000 |
| **Total Consumption** | **€299,000** |

*Table 17: Overall Total Consumption*

The resulting total consumption of €299,000 represents approximately 84.23% of the revised budget of €355,000.

This measure provides a more comprehensive view of budget utilization than actual costs alone.

## 16.6 Available Budget

Available budget represents the remaining amount after deducting total consumption from the revised budget.

**Formula:**

Available Budget = Revised Budget − Total Consumption

For the overall simulated company:

€355,000 − €299,000 = €56,000

| Item | Amount |
|---|---|
| Revised Budget | €355,000 |
| Total Consumption | €299,000 |
| **Available Budget** | **€56,000** |

*Table 18: Overall Available Budget Calculation*

The total available budget is therefore €56,000.

At departmental level, available budget helps identify which cost centers retain sufficient budget headroom and which may require closer monitoring.

A positive available budget does not automatically mean that further expenditure should be approved. Management should also consider upcoming operational requirements, outstanding commitments, and internal approval procedures.

## 16.7 Departmental Budget Consumption Results

The departmental analysis shows differences in budget utilization across the five cost centers.

| Department | Revised Budget | Actual Costs | Open Commitments | Total Consumption | Available Budget |
|---|---|---|---|---|---|
| Production | €138,000 | €75,000 | €30,000 | €105,000 | €33,000 |
| Warehouse | €55,000 | €30,000 | €17,000 | €47,000 | €8,000 |
| Procurement | €32,000 | €20,000 | €12,000 | €32,000 | €0 |
| Quality | €50,000 | €30,000 | €15,000 | €45,000 | €5,000 |
| Logistics | €80,000 | €45,000 | €25,000 | €70,000 | €10,000 |
| **Total** | **€355,000** | **€200,000** | **€99,000** | **€299,000** | **€56,000** |

*Table 19: Departmental Budget Consumption Results*

Production has the highest revised budget and the greatest remaining budget availability. Procurement has reached its revised budget limit, while Quality, Warehouse, and Logistics have comparatively limited remaining balances.

These results demonstrate why budget monitoring should be performed at both company-wide and departmental levels.

## 16.8 Interpretation of Results

The analysis indicates that overall budget utilization is 84.23%. Although the company retains €56,000 in available budget, this aggregate figure does not fully reflect the financial position of each department.

Procurement has consumed or committed its entire revised budget of €32,000. This department should therefore receive priority attention when reviewing additional purchasing requirements.

Quality has €5,000 remaining, while Warehouse has €8,000 and Logistics has €10,000. These departments may require closer monitoring to ensure that expected expenditure remains within available allocations.

Production has €33,000 available and utilization of approximately 76.09%. Compared with the other departments, Production has the greatest remaining budget headroom.

The results illustrate the importance of analyzing actual costs, open commitments, and available budget together when assessing departmental financial performance.

All figures and interpretations in this section relate to the simulated portfolio dataset and are not actual financial results from ABC Manufacturing GmbH.

# 17. Budget Utilization and Budget Availability Control Analysis

## 17.1 Introduction

Budget utilization indicates the proportion of a department's revised budget that has been consumed or committed. It provides an important indicator for monitoring expenditure and identifying departments that may be approaching their budget limits.

In this project, utilization is calculated using total consumption, which includes actual costs and open commitments. The revised budget is used as the reference value.

## 17.2 Budget Utilization Formula

Budget Utilization (%) = (Total Consumption ÷ Revised Budget) × 100

This calculation provides a percentage that can be compared across departments with different budget sizes.

A higher utilization percentage indicates that a larger proportion of the available budget has been consumed or committed. A utilization percentage of 100% indicates that total consumption equals the revised budget.

## 17.3 Departmental Budget Utilization Results

| Department | Revised Budget | Total Consumption | Utilization |
|---|---|---|---|
| Production | €138,000 | €105,000 | 76.09% |
| Warehouse | €55,000 | €47,000 | 85.45% |
| Procurement | €32,000 | €32,000 | 100.00% |
| Quality | €50,000 | €45,000 | 90.00% |
| Logistics | €80,000 | €70,000 | 87.50% |
| **Overall** | **€355,000** | **€299,000** | **84.23%** |

*Table 20: Departmental Budget Utilization Results*

The results show that Procurement has the highest utilization at 100%, followed by Quality at 90%. Logistics and Warehouse also have relatively high utilization, while Production has the lowest utilization at approximately 76.09%.

## 17.4 Budget Availability Control (BAC) Classification

Budget Availability Control is represented in this portfolio project through illustrative utilization thresholds. These rules are used to classify the budget position of each department.

The thresholds applied are:

| Utilization | Classification | Interpretation |
|---|---|---|
| Below 80% | NORMAL | Budget consumption remains below the warning threshold. |
| From 80% to below 100% | WARNING | The department is approaching its budget limit and requires monitoring. |
| 100% or above | ERROR | The budget limit has been reached or exceeded. |

*Table 21: BAC Classification Thresholds*

These are project-specific analytical rules. They are not intended to represent a universal SAP configuration or standard threshold applicable to every organization.

## 17.5 Departmental BAC Results

| Department | Utilization | BAC Status |
|---|---|---|
| Production | 76.09% | NORMAL |
| Warehouse | 85.45% | WARNING |
| Procurement | 100.00% | ERROR |
| Quality | 90.00% | WARNING |
| Logistics | 87.50% | WARNING |

*Table 22: Departmental BAC Results*

Production is classified as NORMAL because utilization is below 80%.

Warehouse, Quality, and Logistics are classified as WARNING because their utilization is at least 80% but below 100%.

Procurement is classified as ERROR because its consumption has reached 100% of its revised budget.

## 17.6 Interpretation and Management Implications

The BAC classification helps prioritize management attention.

Procurement should receive immediate attention because no budget remains available under the current calculation. Any additional expenditure should be reviewed against the organization's approval procedures and applicable budget controls.

Quality, Logistics, and Warehouse should be monitored because their utilization exceeds the illustrative warning threshold. Management should review expected expenditure and outstanding commitments to determine whether corrective action is necessary.

Production retains the greatest budget headroom and remains below the warning threshold.

This classification provides a simple way to communicate financial risk through the dashboard. However, the classification is based on simulated data and project-defined thresholds rather than a live SAP control mechanism.

# 18. Power BI Dashboard Development

## 18.1 Objective

Microsoft Power BI was used to transform the structured financial data into a visual dashboard for budget monitoring and departmental comparison.

The dashboard brings together the main financial indicators and provides a consolidated view of revised budgets, actual costs, open commitments, budget availability, utilization, and BAC status.

The dashboard was developed using data prepared in Microsoft Excel. No live SAP system connection was used.

## 18.2 Data Import and Preparation

The Excel workbook contains the structured datasets used in the analysis, including departmental information, budgets, actual costs, commitments, adjustments, and final analytical results.

The relevant worksheets were imported into Power BI Desktop.

The data fields were then used to build KPI cards and departmental charts. Numeric fields were formatted for financial or percentage display where appropriate.

The dashboard uses the prepared analytical results as its reporting foundation.

## 18.3 KPI Cards

Four KPI cards were created to present the company's overall financial position.

**Total Revised Budget**

This KPI displays the total revised budget of €355,000 after the simulated budget adjustments.

**Total Actual Costs**

This KPI displays actual costs of €200,000 across the five departments.

**Open Commitments**

This KPI displays outstanding commitments of €99,000.

**Available Budget**

This KPI displays the remaining budget of €56,000 after subtracting actual costs and open commitments from the revised budget.

| KPI Card | Value |
|---|---|
| Total Revised Budget | €355,000 |
| Total Actual Costs | €200,000 |
| Open Commitments | €99,000 |
| Available Budget | €56,000 |

*Table 23: Power BI Dashboard KPI Cards*

Together, these four cards provide a quick overview of the financial position without requiring the user to inspect each department individually.

## 18.4 Budget Utilization by Department

The budget utilization chart compares the percentage of revised budget consumed or committed by each department.

The chart highlights Procurement as the department with the highest utilization at 100%. Quality follows at 90%, Logistics at 87.50%, and Warehouse at 85.45%.

Production has the lowest utilization at 76.09%.

| Department | Budget Utilization |
|---|---|
| Procurement | 100.00% |
| Quality | 90.00% |
| Logistics | 87.50% |
| Warehouse | 85.45% |
| Production | 76.09% |

*Table 24: Budget Utilization by Department (Dashboard Chart)*

This visual helps users identify departments that may require closer monitoring.

## 18.5 Revised Budget versus Total Consumption

The comparison chart presents revised budget and total consumption for each department.

The chart allows users to compare the allocated budget with actual costs plus open commitments.

It also makes differences in budget headroom visible. For example, Production has €33,000 remaining, while Procurement has no remaining budget under the project's calculation.

| Department | Revised Budget | Total Consumption |
|---|---|---|
| Production | €138,000 | €105,000 |
| Warehouse | €55,000 | €47,000 |
| Procurement | €32,000 | €32,000 |
| Quality | €50,000 | €45,000 |
| Logistics | €80,000 | €70,000 |

*Table 25: Revised Budget versus Total Consumption (Dashboard Chart)*

## 18.6 Available Budget by Department

This chart compares the remaining budget available in each department.

The results show that Production has the largest available budget at €33,000. Logistics has €10,000 available, Warehouse has €8,000, and Quality has €5,000. Procurement has no available budget remaining.

| Department | Available Budget |
|---|---|
| Production | €33,000 |
| Logistics | €10,000 |
| Warehouse | €8,000 |
| Quality | €5,000 |
| Procurement | €0 |

*Table 26: Available Budget by Department (Dashboard Chart)*

This visualization supports the identification of departments with limited budget flexibility.

## 18.7 BAC Status by Department

The BAC status visual communicates the classification assigned to each department.

Production is classified as NORMAL. Warehouse, Quality, and Logistics are classified as WARNING. Procurement is classified as ERROR.

| Department | BAC Status |
|---|---|
| Production | NORMAL |
| Warehouse | WARNING |
| Procurement | ERROR |
| Quality | WARNING |
| Logistics | WARNING |

*Table 27: BAC Status by Department (Dashboard Visual)*

The visual provides a concise view of which departments fall within the normal range and which require closer attention under the project's illustrative thresholds.

## 18.8 Dashboard Design and Usability

The dashboard was organized to present headline KPIs before the more detailed departmental charts.

The KPI cards provide an overview of the total financial position, while the charts below provide departmental detail.

Consistent titles, readable labels, appropriate number formatting, and clear status categories help make the report easier to interpret.

The dashboard is intended as a management-reporting demonstration rather than a replacement for formal financial controls or an operational SAP system.

# 19. Key Findings and Recommendations

## 19.1 Summary of Findings

The analysis identified several important findings:

| Key Finding | Result |
|---|---|
| Revised company budget | €355,000 |
| Total actual costs | €200,000 |
| Open commitments | €99,000 |
| Total consumption | €299,000 |
| Remaining budget | €56,000 |
| Overall utilization | Approx. 84.23% |
| Procurement utilization | 100% |
| Production utilization (lowest) | Approx. 76.09% |

*Table 28: Summary of Key Findings*

These findings demonstrate that an overall positive budget position can coexist with individual departments that have exhausted their available budgets.

## 19.2 Recommendations for Procurement

Procurement should be prioritized for budget review because its total consumption equals its revised budget.

Management should review pending purchasing requirements, open commitments, and expected additional costs before authorizing further expenditure.

Where necessary, purchasing plans should be reprioritized or additional budget approval should be considered through the appropriate internal process.

## 19.3 Recommendations for Quality

Quality has €5,000 remaining, corresponding to 90% utilization.

The department should monitor expected expenditure and outstanding commitments to avoid exceeding its revised budget.

### 19.4 Recommendations for Logistics and Warehouse

Logistics has €10,000 remaining, while Warehouse has €8,000 remaining.

Both departments are classified as WARNING under the project's illustrative thresholds. Management should review upcoming operational expenditure and monitor changes in commitments.

### 19.5 Recommendations for Production

Production has €33,000 remaining and utilization of 76.09%.

Routine monitoring should continue to ensure that expenditure remains aligned with operational needs and the revised budget.

The available budget should not automatically be interpreted as surplus funds that can be transferred or spent without authorization.

### 19.6 General Recommendations

For effective budget monitoring, organizations should:

- Review actual costs and open commitments regularly.
- Monitor departments approaching their budget limits.
- Investigate significant differences between budget and consumption.
- Maintain accurate and current budget data.
- Use consistent reporting definitions.
- Ensure that budget changes follow the appropriate approval procedures.
- Validate analytical results before using them for management decisions.

These recommendations are based on the simulated case study and would need to be adapted to an organization's actual financial policies and systems.

## 20. Limitations and Assumptions

### 20.1 Simulated Data

All financial information used in this project is simulated. The company, departments, budgets, costs, commitments, and adjustments are fictional and are used solely for educational and portfolio purposes.

### 20.2 No Live SAP Integration

The project does not connect to a live SAP S/4HANA environment. The Excel workbook and Power BI dashboard demonstrate an analytical workflow inspired by SAP Controlling concepts.

No live SAP transactions, configuration changes, or financial postings were performed.

### 20.3 Illustrative BAC Thresholds

The NORMAL, WARNING, and ERROR classifications use project-specific utilization thresholds.

Actual organizations may use different thresholds, control procedures, or SAP configuration settings depending on their business requirements.

### 20.4 Reporting Scope

The analysis focuses on departmental budgets, actual costs, open commitments, consumption, and available budget.

It does not include detailed cost-element analysis, monthly spending trends, forecasting, currency conversion, or transaction-level reconciliation.

### 20.5 Interpretation of Commitments

For this project, open commitments are added to actual costs to calculate total consumption.

In a real financial system, the precise treatment of commitments depends on the relevant reporting definitions and system configuration. Care must be taken to avoid double counting when actual costs and commitments are extracted from operational data.

## 21. Conclusion

This project demonstrates how structured financial data can be used to build a departmental budget monitoring and management-reporting solution.

The Excel workbook provides the data structure and calculations, while the Power BI dashboard communicates the results through KPI cards and departmental visualizations.

The analysis shows that total consumption amounts to €299,000 against a revised budget of €355,000, leaving €56,000 available. Overall utilization is approximately 84.23%.

At departmental level, Procurement has reached its budget limit, while Quality, Logistics, and Warehouse fall within the illustrative warning range. Production retains the greatest budget headroom.

The project demonstrates practical application of cost center monitoring, budget analysis, commitment tracking, KPI calculation, and dashboard development.

It also provides a foundation for further learning in SAP S/4HANA Controlling, financial analytics, and Power BI.

The work should be understood as a simulated portfolio project rather than a live SAP implementation or professional financial audit.

# 22. Future Improvements

Several improvements could extend the project.

## 22.1 Monthly Budget Monitoring

Adding monthly data would allow users to examine changes in spending and utilization over time.

## 22.2 Forecasting

Forecasting methods could be introduced to estimate expected year-end costs and identify potential budget overruns before they occur.

## 22.3 Detailed Cost Analysis

The project could be extended to include cost categories, cost elements, vendors, purchase orders, and transaction-level data.

## 22.4 Automated Data Refresh

A future version could use a supported data connection to refresh reporting data automatically, subject to appropriate access and data governance controls.

## 22.5 SAP Training Environment Integration

The project could be reproduced or extended using data from an authorized SAP training environment. This would provide an opportunity to practice working with real SAP screens, reports, and system-generated data.

## 22.6 Advanced Power BI Features

Additional features could include drill-through pages, interactive department filters, trend analysis, variance indicators, and detailed departmental reports.

These improvements would expand the project's analytical capabilities and provide further opportunities to demonstrate financial reporting and business intelligence skills.